Stochastic Processes – AY 2022/2023
written test – July 03, 2023 – part A (90 minutes)

E1 Consider a Go-Back-N protocol over a two-state Markov channel, where the average number of consecutive good slots is 100 and the average fraction of bad slots is 0.1. The packet error probability is 1 for a bad slot and 0 for a good slot, respectively. The round-trip time is $m = 2$ slots, i.e., a packet that is erroneous in slot t will be retransmitted in slot $t + 2$.

(a) Compute the throughput that could be obtained if packets were directly transmitted over the channel without using any protocol

(b) Compute the throughput of the Go-Back-N protocol for an error-free feedback channel

(c) Compute the throughput of the Go-Back-N protocol for a feedback channel subject to iid errors with probability 0.1.

E2 Consider a Markov chain X_n with the following transition matrix (states are numbered from 0 to 2):

$$P = \begin{pmatrix} 0.2 & 0.4 & 0.4 \\ 0.4 & 0.2 & 0.4 \\ 0 & 0 & 1 \end{pmatrix}$$

(a) Draw the transition diagram, and find the probability distribution of X_1, X_2 and X_{1000}, given $X_0 = 0$

(b) Let $W_{ij}^{(n)} = E\left[\sum_{k=0}^{n-1} I\{X_k = j\} \middle| X_0 = i\right]$ be the average number of visits to state j during the first n time slots, given that the chain starts in state i. Compute $\lim_{n\to\infty} W_{0j}^{(n)}$ for $j = 0, 1, 2$.

(c) Compute the average duration of the transient evolution of the chain, i.e., the time index at which the chain is absorbed, given $X_0 = i$ for $i = 0, 1$.

E3 Consider two independent Poisson processes $X_1(t)$ e $X_2(t)$, where $X_i(t)$ is the number of arrivals for process i during $[0, t]$. The average number of arrivals per unit time of the two processes is $\lambda_1 = 0.5$ and $\lambda_2 = 1$, respectively.

(a) Compute $P[X_1(1) = 1 | X_1(1) + X_2(1) = 3]$ and $P[X_1(1) + X_2(1) = 3 | X_1(1) = 1]$

(b) Compute $P[X_2(1) = 1 | X_1(2) + X_2(2) = 3]$ and $P[X_1(2) + X_2(2) = 3 | X_2(1) = 1]$

E4 Consider a network node able to handle traffic at 20 Gbps under normal conditions. The node is subject to attacks, that arrive according to a Poisson process of rate $\lambda = 1/T_0$. For each attack, the node has a probability $1 - \alpha$ of being infected, whereas with probability α the attack has no consequence. When a node gets infected, it automatically starts a clean-up process that lasts T_1 and occupies 50% of its resources, so that during this phase the node can only handle 10 Gbps. The clean-up process is successful with probability β (in which case the node starts working normally), whereas with probability $1 - \beta$ it fails and the node needs to be restored manually by a human operator, which takes T_2, during which time the node does not handle any traffic. After being manually restored, the node starts working normally.

(a) By identifying an appropriate renewal cycle, compute the fraction of the time the node is not handling any traffic

(b) Compute the average traffic per unit time (in Gbps) handled by the node,

(c) Compute how often (e.g., how many times a day on average) a human operator's intervention is needed.

(For all the above quantities, find mathematical expressions as a function of the parameters, and then compute their numerical values for $T_0 = 10$ minutes, $T_1 = 10$ minutes, $T_2 = 3$ hours, $\alpha = 0.99$, $\beta = 0.9$.)